🧠 Multithreading Assignments (C#)

🟢 Assignment 1: Parallel Number Processing

🎯 **Scenario:**

You are building a system that processes a list of numbers concurrently to improve performance.

🧩 **Requirements:**

1. Create a list of numbers from 1 to 50
2. Split the list into 5 parts
3. Create **5 threads**, each processing one part

👉 **Each thread should:**

- Print numbers assigned to it
- Calculate the sum of its numbers

⚠️ **Constraints:**

- Use Thread class only
- Use Thread.Start() and Thread.Join()
- Store partial results and compute **final sum in main thread**

🎯 **Expected Output:**

Thread 1 Sum: XX

Thread 2 Sum: XX

...

Final Sum: XXXX

💡 **Concepts Covered:**

- Thread creation
- Thread lifecycle (Start, Join)

- Parallel execution

🟡 **Assignment 2: Bank Account (Synchronization Problem)**

🎯 **Scenario:**

Simulate a bank account where multiple users withdraw money at the same time.

🧩 **Requirements:**

1. Create a BankAccount class:

Balance

Withdraw(amount)

2. Create **3 threads**, each trying to withdraw money

⚠️ **Problem:**

Without synchronization, balance may become incorrect.

👉 **Tasks:**

1. First run WITHOUT synchronization → observe issue
2. Then fix using:
    - lock keyword (preferred)

🎯 **Expected Behavior:**

- Balance should never go negative
- Proper deduction should happen